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| Ex No – 7 | IPC Using Shared Memory |
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**Program :**

**sender.c**

```
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

#define SharedMemSize 50

void main()
{
    char c;
    int shmid;
    key_t key;
    char *shared_memory;

    key = 5677;

    if ((shmid = shmget(key, SharedMemSize, IPC_CREAT | 0666))
    < 0)
    {
        perror("shmget");
        exit(1);
    }
```

```

if ((shared_memory = shmat(shmid, NULL, 0)) == (char *) -1)
{
    perror("shmat");
    exit(1);
}

sprintf(shared_memory, " Welcome to Shared Memory");

sleep(2);

exit(0);
}

```

### **receiver.c**

```

#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>

#define SharedMemSize 50

void main()
{
    int shmid;
    key_t key;
    char *shared_memory;

    key = 5677;

    if ((shmid = shmget(key, SharedMemSize, 0666)) < 0)
    {
        perror("shmget");
        exit(1);
    }

    if ((shared_memory = shmat(shmid, NULL, 0)) == (char *) -1)

```

```
{  
    perror("shmat");  
    exit(1);  
}  
  
printf("Message Received: %s \n", shared_memory);  
  
exit(0);
```

**Output :**

```
[cse32@localhost ~]$ vi ipc_sender.c  
[cse32@localhost ~]$ vi ipc_receiver.c  
[cse32@localhost ~]$ gcc ipc_sender.c -o ipc_sender  
[cse32@localhost ~]$ ./ipc_sender  
[cse32@localhost ~]$ gcc ipc_receiver.c -o ipc_receiver  
[cse32@localhost ~]$ ./ipc_receiver  
Message Received: Welcome to Shared Memory  
[cse32@localhost ~]$
```